

INNER WHALEBACKS, NT, Canada

WMO: 711620

Lat: 61.921N

Lon: 113.726W

Elev: 542

StdP: 14.41

Time Zone: -7.00 (NAM)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.0	-28.0	-39.4	0.6	-32.0	-35.0	0.8	-27.9	35.5	-2.9	32.1	-5.2	12.0	310	0.980

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.5	68.6	62.7	65.9	60.8	63.7	59.2	63.5	67.8	61.4	65.3	59.6	63.3	9.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.5	83.3	66.6	59.3	77.0	64.5	57.5	71.9	62.6	29.2	67.7	27.7	65.6	26.4	63.4	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.9	30.0	27.2	-31.0	74.8	4.7	3.6	-34.4	77.4	-37.2	79.5	-39.8	81.5	-43.3	84.2		
(5)			-31.8	66.6	4.6	4.4	-35.1	69.8	-37.8	72.4	-40.4	74.8	-43.7	78.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.8	-8.6	-5.9	2.1	19.8	34.8	46.2	56.4	57.2	48.4	35.6	19.5	2.4		
	DBStd	24.85	12.51	11.76	12.94	10.95	6.04	4.93	6.16	4.82	6.01	6.96	9.51	12.18		
	HDD50	9328	1815	1566	1485	905	470	136	12	4	99	447	914	1475		
	HDD65	14303	2280	1986	1950	1355	935	565	273	245	498	912	1364	1940		
	CDD50	509	0	0	0	0	0	20	211	227	51	0	0	0		
	CDD65	10	0	0	0	0	0	0	7	3	0	0	0	0		
	CDH74	7	0	0	0	0	0	0	7	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	14.5	14.0	13.4	14.2	14.4	13.0	13.4	12.2	13.4	15.8	17.3	17.4	15.1		
	PrecAvg	13.5	0.7	0.5	0.6	0.5	0.9	1.5	2.0	2.2	1.7	1.3	1.1	0.7		
(15) Precipitation	PrecMax	17.6	1.2	1.1	1.1	0.7	1.7	3.7	3.9	3.7	3.2	2.4	1.8	1.2		
	PrecMin	9.1	0.2	0.1	0.2	0.2	0.1	0.2	0.7	0.9	0.5	0.5	0.6	0.3		
	PrecStd	1.6	0.3	0.2	0.3	0.2	0.4	0.8	0.7	0.7	0.7	0.4	0.3	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.7	24.4	33.9	41.2	52.6	65.0	74.6	70.7	66.3	51.1	41.6	30.6		
		MCWB	27.1	22.8	31.7	37.6	44.7	54.5	67.2	63.5	65.6	47.5	38.7	29.3		
	2%	DB	19.3	19.8	30.1	38.4	46.7	59.2	70.3	67.3	58.9	48.7	37.3	27.3		
		MCWB	18.4	19.0	28.5	35.6	41.2	51.7	64.1	61.6	55.1	45.7	35.2	26.1		
	5%	DB	12.9	14.6	26.5	36.8	43.7	55.4	67.5	65.1	57.1	46.6	34.9	23.7		
		MCWB	12.3	13.8	25.3	34.4	39.2	49.8	61.9	60.4	53.9	44.0	33.3	22.8		
	10%	DB	8.2	10.7	21.7	35.1	41.5	52.4	64.9	63.4	55.5	44.2	32.1	19.8		
		MCWB	7.6	10.0	20.4	33.2	38.0	47.6	60.3	59.1	52.6	41.7	30.8	19.2		
	0.4%	WB	27.1	22.9	31.7	37.8	45.1	55.9	68.1	64.4	65.9	48.1	38.8	29.4		
		MCDB	27.6	24.2	33.3	40.6	51.2	62.5	73.2	69.1	66.5	50.2	41.6	30.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	18.4	19.0	28.5	35.8	41.7	52.7	64.8	62.4	55.8	46.1	35.5	26.3		
		MCDB	19.1	19.9	30.0	38.0	45.8	58.1	69.5	66.2	58.1	48.3	37.1	27.2		
	5%	WB	12.3	14.0	25.3	34.5	39.6	50.3	62.5	60.9	54.3	44.1	33.4	22.8		
		MCDB	12.8	14.5	26.6	36.4	43.2	54.7	66.6	64.7	56.7	46.5	34.9	23.6		
	10%	WB	7.6	10.0	20.5	33.3	38.1	48.1	60.6	59.4	52.9	41.8	30.8	19.1		
		MCDB	8.1	10.6	21.7	35.1	41.1	52.1	64.5	63.0	55.3	44.1	32.0	19.8		
	5% DB	MDBR	10.0	11.8	16.0	16.7	11.6	11.3	10.0	6.5	5.8	5.7	7.4	9.4		
		MCDBR	12.3	14.3	17.4	14.7	13.5	16.2	12.4	8.6	7.0	6.6	8.4	11.9		
		MCWBR	12.1	14.2	16.3	12.6	9.4	10.7	7.9	5.8	5.1	5.9	7.9	11.5		
		MCWBR	12.2	14.5	17.2	13.7	13.0	15.2	11.2	8.3	6.7	6.5	8.3	11.5		
(35) Mean Daily Temperature Range	5% WB	MCWBR	12.1	14.3	16.2	11.9	9.3	10.9	7.9	5.9	5.3	6.0	7.9	11.1		
	taub		0.193	0.243	0.267	0.310	0.321	0.348	0.372	0.342	0.314	0.274	0.198	0.154		
		taud	2.066	2.225	2.262	2.321	2.491	2.424	2.412	2.500	2.544	2.425	2.067	1.937		
(40) Clear-Sky Solar Irradiance	Ebn at Noon		171	237	272	280	286	279	267	268	256	225	162	127		
		Edh at Noon	11	19	27	32	30	32	32	27	22	16	10	6		
	RadAvg		94	353	909	1547	1870	1984	1763	1325	800	321	94	44		
(44) All-Sky Solar Radiation	RadStd		10	28	58	82	117	159	86	64	60	32	13	4		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)		N/A	N/A	+2.54	N/A	N/A	+0.86	N/A	N/A	N/A

Nomenclature: See separate page